भारतीय प्रौद्योगिकी संस्थान कानपुर
Indian Institute of Technology Kanpur

DEPARTMENT OF

Sustainable Energy Engineering

Postgraduate Programmes

M.Tech • MS by Research • PhD • Integrated PhD





www.iitk.ac.in/see

ABOUT THE DEPARTMENT

Energy sustainability, from materials to policy

The Department of Sustainable Energy Engineering (SEE) was established at IIT Kanpur in December 2020 to educate engineers and researchers to support India's energy transition. Our first postgraduate students joined in July 2021.

Our teaching and research cover the whole energy chain, from how clean energy is generated to how it is stored, distributed, and used. We also work on how carbon dioxide is captured and converted, how critical minerals are recovered and recycled, and how the energy transition is modelled and governed.

Our faculty members come from science, engineering, and policy backgrounds, and several hold joint appointments with the Departments of Aerospace Engineering and Civil Engineering. Our programmes aim to train engineers who can connect the science of energy materials and processes to the design of real energy systems.

2020
the year the department was founded

16
faculty members, including joint appointments

5
research verticals

4
regular postgraduate programmes



Faculty, staff, and students of the department.

International links

Through the RICE-IITK Collaborative Center, we work with Rice University, USA, on joint research. Students can be co-advised by faculty at both institutions and participate in exchanges (**iitk.ac.in/see/rice-iitk**). IIT Kanpur also offers joint degree programmes with several universities abroad (**iitk.ac.in/oir/joint-degree-programs**).

Research centres

Our research is supported by these centres:

- Chandrakanta Kesavan Centre
- DST-ICMAP Centre
- EIL Centre
- DIA Centre of Excellence on Batteries
- IITK-JSW Cement CCUS Centre
- UPNEDA Centre of Excellence for Green Hydrogen

RESEARCH

Five research verticals

The five verticals of the department follow energy from production to the end of its life. They overlap, and many of our faculty members work in more than one. Our research ranges from basic science to working prototypes.

01 Energy generation

Solar photovoltaics · Solar thermal · Wind energy · Fuel cells · Green hydrogen

We develop new photovoltaic materials, including perovskite, organic, and tandem solar cells, and work on solar thermal and thermochemical systems, wind energy and aerodynamics, fuel cells, and green hydrogen.

FACULTY

Ashish Garg, Srinivas Karthik Yadavalli, Prabodh Bajpai, Sayan Kar, Soumyabrata Roy, Laltu Chandra, Vaibhav Arghode, Debopam Das, Lalit M. Pant, Amarendra Edpuganti, Vivek Verma

02 Energy storage

Batteries · Supercapacitors · Solar thermal · Carbon capture, utilization and storage

Our work on storage covers solid-state and next-generation batteries, supercapacitors, the modeling of electrochemical storage, and solar thermal storage. It also includes capturing carbon dioxide for storage or use.

FACULTY

Sudarshan Narayanan, Vivek Verma, Soumyabrata Roy, Laltu Chandra, Amarendra Edpuganti, Lalit M. Pant, Sayan Kar, Sheo Shankar Rai

03 Energy distribution

Smart grid and microgrid · Power distribution · Electric mobility

Faculty in this vertical work on hybrid AC-DC microgrids, smart grids, bringing renewable power into the grid and electricity markets, as well as power electronics for electric and fuel cell vehicles.

FACULTY

Prabodh Bajpai, Amarendra Edpuganti, Swathi Battula, Gururaj Vishwanath

04 Energy modeling and policy

Energy policy · Climate modeling · Decarbonization · Net zero

We study energy and climate policy and regulation, climate modeling and air pollution, and ways to decarbonize industry. This vertical also covers net-zero buildings, from energy metering to occupant behaviour and indoor air quality.

FACULTY

Deepika Swami, Sheo Shankar Rai, Aakash C. Rai, Ashish Garg, Sachchida Nand Tripathi

05 Critical materials and recycling

Critical minerals · Urban mining · E-waste recycling · Circular economy · Materials for clean energy

Clean energy depends on critical minerals such as lithium, cobalt, nickel, and the rare earths. We work on securing their supply, recovering them from electronic, battery, and solar panel waste, and developing the materials and processes of a circular economy.

FACULTY

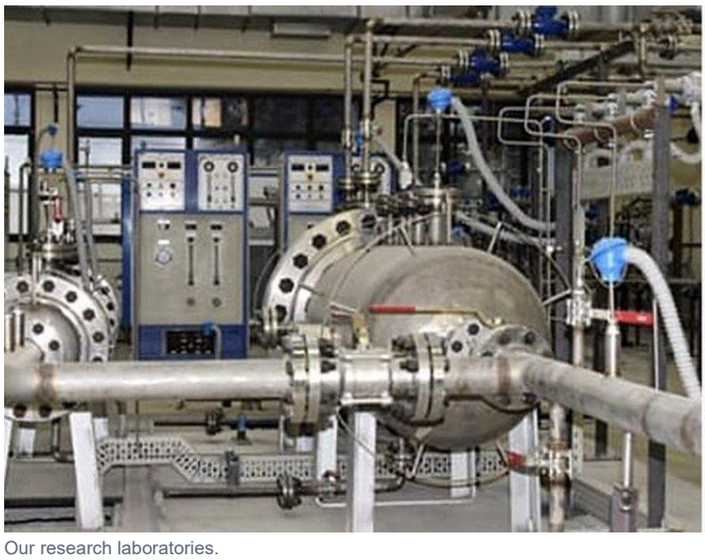
Vivek Verma, Sheo Shankar Rai, Ashish Garg, Srinivas Karthik Yadavalli, Sayan Kar

RESEARCH AREAS ACROSS THE DEPARTMENT

Batteries · Building design · Carbon capture and utilization · Critical materials and recycling · Electric vehicles · Energy economics and policy · Energy modeling · Energy efficiency · Fuel cells · Hydrogen and alternative fuels · Microgrids · Net zero and carbon neutrality · Smart grid and renewables integration · Solar photovoltaics · Solar thermal generation and storage · Water · Wind energy

ACILITIES

Laboratories and instruments




Our research laboratories.

Teaching laboratory

Every postgraduate student takes SEE605, which includes our Sustainable Energy Technologies laboratory. There, students get hands-on training in solar photovoltaics, solar thermal systems, energy storage, hydrogen and fuel cells, smart grids, wind energy, basic electronics, temperature and flow measurement, and materials synthesis and characterization.

Institute facilities

Students can also use the institute's central facilities, including:

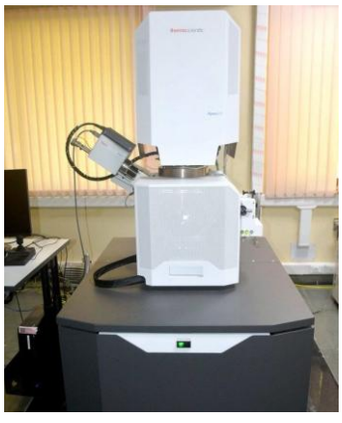
- Advanced Centre for Materials Science
- Advanced Imaging Centre
- Centre for Nanosciences

Research laboratories

Our research laboratories include:

- Solar photovoltaics fabrication laboratory
- Battery materials, cell development, and characterization laboratory
- Hydrogen generation and storage laboratories
- Materials and device laboratory
- Clean room and characterization facility
- Sustainable Power Innovation Technologies (NOVEL) laboratory
- Solar thermal systems laboratory
- Thermal-fluids laboratory

Central instruments in the department



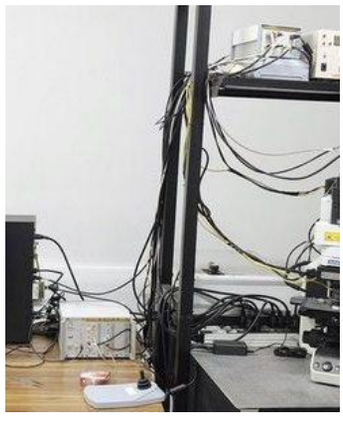
Field emission SEM
Thermo Fisher Apreo 2C



X-ray diffractometer
Rigaku SmartLab SE



Surface area and porosity analyzer
Anton Paar Autosorb 6100



Time-resolved photoluminescence
PicoQuant FluoTime 300

The department also has a stylus profilometer, a UV-visible-NIR spectrophotometer, a photoluminescence spectrometer, LC-MS, GC and GC-MS, and gel permeation chromatography.

FACULTY

Department faculty



Prof. Aakash C. Rai
Energy-efficient buildings, impact of climate change on heating and cooling of buildings, indoor air quality, airborne infection transmission, and air pollution
aakashrai@iitk.ac.in



Prof. Amarendra Edpuganti
Power electronics applications in renewable energy, electric vehicles, and fuel cell vehicles
amarendrae@iitk.ac.in



Prof. Ashish Garg
HEAD OF DEPARTMENT
Solar photovoltaics, recycling, materials, and devices, decarbonization
ashishg@iitk.ac.in



Prof. Debopam Das
JOINT WITH AEROSPACE ENGINEERING
Wind energy, computational fluid dynamics, UAVs
das@iitk.ac.in



Prof. Deepika Swami
Energy policy and climate change
dswami@iitk.ac.in



Prof. Lalit M. Pant
Electrolyzers and fuel cell design, electrochemical energy conversion and storage, numerical modeling, porous media transport
lalit@iitk.ac.in



Prof. Laltu Chandra
Heat transfer and fluid flow, computation and experiment, turbulent flow simulation and modeling, solar thermal sub-system design, nuclear reactor thermal hydraulics
lchandra@iitk.ac.in



Prof. Prabodh Bajpai
Hybrid AC-DC microgrids, smart grid and renewable integration, solar photovoltaics, electricity markets, power system analysis and control
pbajpai@iitk.ac.in



Prof. Sachchida Nand Tripathi
JOINT WITH CIVIL ENGINEERING, ON LEAVE
Climate issues, climate modeling, environment, and air pollution
snt@iitk.ac.in



Prof. Sayan Kar
Carbon capture, CO_2 utilization, energy materials
sayank@iitk.ac.in



Prof. Sheo Shankar Rai
PROFESSOR OF PRACTICE
Critical minerals for the energy transition, sustainable development of mineral resources and mining automation, carbon capture and sequestration (CCS), energy and mining policy
sheo@iitk.ac.in



Prof. Soumyabrata Roy
Carbon capture and utilization, functional materials, processes, and circularity
soumyar@iitk.ac.in



Prof. Srinivas Karthik Yadavalli
Halide perovskites, solar cells, multi-junction photoelectrodes for green hydrogen, photovoltaic recycling
srinivask@iitk.ac.in



Prof. Sudarshan Narayanan
Solid-state batteries, thin films for energy conversion, and advanced characterization
sudarshan@iitk.ac.in



Prof. Vaibhav Arghode
JOINT WITH AEROSPACE ENGINEERING
Solar thermal energy
varghode@iitk.ac.in



Prof. Vivek Verma
E-waste material recovery and recycling, beyond Li-ion battery materials, bulk X-ray characterization (EXAFS, XRF, and XRD)
vermav@iitk.ac.in

FACULTY

Adjunct and visiting faculty

Adjunct faculty



Prof. Anubha Goel

Vehicular emission analysis, air quality assessment, particle and organic pollutant profiling, health risk evaluation, solid waste management, agricultural impacts on climate change

anubha@iitk.ac.in



Prof. Ashutosh Sharma

Science and technology policy, battery materials, sensors, nanofabrication, functional nanomaterials, MEMS/NEMS, soft interfaces, carbon structures

ashutos@iitk.ac.in



Prof. Gururaj Mirle Vishwanath

Renewable energy integration, microgrids, EV-grid interaction, ML applications to power systems

gururajmv@iitk.ac.in



Prof. Himanshu Sharma

Carbon capture, alternative fuels

sharmah@iitk.ac.in



Prof. Swathi Battula

Electricity markets, modeling and design of electrical energy systems, transactive energy, integrated transmission and distribution, energy policy and management

swathi@iitk.ac.in

Visiting faculty



Prof. Sanjay Khare

Sustainable manufacturing, corporate governance, ESG strategy, innovation ecosystems, industry-academia collaboration, TQM, leadership mentorship

skhare@iitk.ac.in



Prof. Satyam Sahay

Advanced engineering and manufacturing technologies, analytics and AI/ML applications, product circularity, and sustainability in metallurgical processes

sahayss@iitk.ac.in

ALUMS

Where SEE alums work

First Solar · Reliance · ReNew · NTPC · SECI · NLC India · HPCL · Hitachi Energy · Mahindra Susten · PwC · Deloitte · Samsung E&A · Jindal Stainless · Caterpillar · Exponent Energy · University of Calgary · Queen's University



Our students presenting their research to faculty and visitors.



Students, faculty, and staff at the department entrance.

CONTACT

Get in touch

For questions about our postgraduate programmes, please write to or call:

DPGC CONVENER
Prof. Lalit M. Pant
dpgc_see@iitk.ac.in
0512-679-2261

DEPARTMENT OFFICE
Department of Sustainable Energy Engineering
office_see@iitk.ac.in
0512-259-2196

JUNIOR ASSISTANT, SEE OFFICE
Mr. Raghvendra Kumar
raghvenk@iitk.ac.in
0512-259-2230

Department website
iitk.ac.in/see

PG programmes
iitk.ac.in/see/pg-programme

How to apply (DoAA)
iitk.ac.in/doaa/admission-procedure

Department of Sustainable Energy Engineering, Indian Institute of Technology Kanpur, Kanpur 208016, Uttar Pradesh, India

Department of Sustainable Energy Engineering

सतत ऊर्जा अभियांत्रिकी विभाग

Indian Institute of Technology Kanpur
Kanpur 208016, Uttar Pradesh, India

www.iitk.ac.in/see